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PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Error Propagation in Multi-Step Agentic AI Workflows for Research Automation...

Measure	Value
Approximate pages	About 8 pages including references
Approximate word count	About 2,600 words
Number of sections	8 main sections + Abstract + Keywords + References
Total number of references	10
Approximate in-text citation occurrences	About 20
Did AI warn you to verify references?	No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	10
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	0
URL only	0
No persistent identifier supplied	0
TOTAL	10

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only

No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	4
R5	5
R6	6
R7	7
R8	8
R9	9
R10	10

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Looks genuine	5	Yes
R2	Looks genuine	5	Yes
R3	Looks genuine	5	Yes
R4	Looks genuine	5	Yes
R5	Looks genuine	5	Yes
R6	Looks genuine	5	Yes
R7	Looks genuine	5	Yes
R8	Looks genuine	5	Yes
R9	Looks genuine	5	Yes
R10	Looks genuine	5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Part F should be completed before Part G. Because this worksheet is being completed after the source-generation step, the entries above are a reconstruction of the initial impression from the unverified bibliography. Replace them with your actual pre-verification answers if you recorded them.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

Persistent identifier: DOI / arXiv / PMID

Publisher or official repository

Google Scholar

Semantic Scholar / OpenAlex

Crossref

PubMed or another discipline-specific database

Exact-title search

Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	Y	Y	Y	Y
R8	Y	Y	Y	Y	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	arXiv 2210.03629 / DOI 10.48550/arXiv.2210.03629	arXiv 2210.03629	Publication exists; title, authors, year and identifier matched.
R2	NeurIPS proceedings / DOI 10.52202/075280-2997	10.52202/075280-2997	Publication exists; title, authors, venue, year and DOI matched.
R3	Microsoft Research / arXiv 2308.08155	arXiv 2308.08155	Publication exists; title, authors, year and arXiv identifier matched.
R4	arXiv 2303.11366	arXiv 2303.11366	Publication exists; title, authors, year and identifier matched.
R5	arXiv 2112.09332	arXiv 2112.09332	Publication exists; title, authors, year and identifier matched.
R6	arXiv 2308.03688	arXiv 2308.03688	Publication exists; title, authors, year and identifier matched.
R7	arXiv 2303.17651	arXiv 2303.17651	Publication exists; title, authors, year and identifier matched.
R8	arXiv 2307.03172	arXiv 2307.03172	Publication exists; title, authors, year and identifier matched.
R9	arXiv 2405.15793 / Princeton publication record	arXiv 2405.15793	Publication exists; title, authors, year and identifier matched.
R10	arXiv 2408.06292	arXiv 2408.06292	Publication exists; title, authors, year and identifier matched.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Publication exists and the main bibliographic details and identifier match.
R2	A	Publication exists and the main bibliographic details and identifier match.

R3	A	Publication exists and the main bibliographic details and identifier match.
R4	A	Publication exists and the main bibliographic details and identifier match.
R5	A	Publication exists and the main bibliographic details and identifier match.
R6	A	Publication exists and the main bibliographic details and identifier match.
R7	A	Publication exists and the main bibliographic details and identifier match.
R8	A	Publication exists and the main bibliographic details and identifier match.
R9	A	Publication exists and the main bibliographic details and identifier match.
R10	A	Publication exists and the main bibliographic details and identifier match.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 10 B = 0 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(10) + 3(0) + 1(0) + 0(0) + 1(0) = 40 points

Authenticity Score = 40 / (4 x 10) x 100 = 100/100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = 10   Total predictions = 10   Prediction Accuracy = 100 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	10
References deeply audited	10
A: Verified	10
B: Wrong metadata	0
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	100 /100
Prediction Accuracy	100 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

No. In this audit, all 10 selected references turned out to be genuine, but they looked convincing before verification. The important point is that their appearance was not enough; each one still had to be checked against a scholarly or publisher record.

2. Did all genuine references actually support the claims for which they were cited?

For the main claims checked in the paper, the genuine references supported the claims for which they were cited. No major claim-citation mismatch was found in the audited sample.

3. What was the most serious citation-integrity problem you observed?

No major fabrication, wrong-metadata case, or identifier mismatch was found in the 10 audited references. The main risk was that a convincing-looking citation could easily be trusted without checking it.

4. What is the single most important lesson you learned from this lab?

The biggest lesson is that a citation should not be trusted just because it looks correct. We need to check both whether the source exists and whether it actually supports the claim.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: ...Chintan Maneesh Gangurde...

Roll Number: ..MHM2026003...

Signature: ..Chintan Maneesh Gangurde...

Date: ..21 August 2026...

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission